

$$8 \cdot 8 \cdot 7 \cdot 5 = 64 \cdot 35$$

$$= 2240$$

$$\begin{array}{r} 64 \cdot 35 \\ 35 \\ \hline 192 \\ 2240 \end{array}$$

2) $\underline{1} \underline{1} \dots \underline{0} \underline{0} \underline{0}$ a) $2^5 = 32$ ✓

b) $\underline{\quad} \underline{\quad} \dots \underline{\quad} \underline{\quad} = 2^9 = 512$ ✓

c) 2^9 ways - if num
 odd $\Rightarrow 1$ else $= 0$
 $\Rightarrow 512$

d) 512 answers
 0 de 1 - 1
 2 de 1 - $P_3 = 3! = 6 \mid \Rightarrow 7105$

e) $\underline{2} \underline{1} \underline{1} \underline{1} = 2 \cdot 1 \cdot 1 \cdot 1 = 2$ $\rightarrow 2$ ntuning ✓

3) 8
 $1, 3, 5, 7$ $\neq$ albe 1 albastru
 $C_8^1 = 8$ moduri

$$C_8^3 = \frac{8!}{3!5!} = \frac{6 \cdot 7 \cdot 8}{2 \cdot 3} = 56$$

$\Rightarrow 112 + 16 = 128$ $C_8^5 = \frac{8!}{5!3!} = 56$
 $C_8^7 = 8$ moduri ✓

4) $\{0, 1, 2, 3, 4, 5, 6, 7\}$

$$\frac{3}{3} \frac{4}{4} \frac{3}{3} \frac{3}{3} \frac{2}{2} = 3^3 \cdot 2^3 = 216$$

$$\frac{4}{4} \frac{4}{4} \frac{3}{3} \frac{3}{3} \frac{2}{2} = 4^2 \cdot 2^5 = 9 \cdot 32 = 288 \quad \Bigg| \Rightarrow \text{So } 4 \text{ moduli}$$

5) $A = \{1, 2, 3\}$ a) $f: A \rightarrow B$

$B = \{1, 2, 3, 4\}$ $4^3 = 64$

c) $g: B \rightarrow A$ b) $\begin{array}{l} 1 \rightarrow 4 \\ 2 \rightarrow 3 \\ 3 \rightarrow 2 \end{array} \Bigg| \Rightarrow 4 \cdot 3 \cdot 2 = 24$

$3^4 = 81$

d) Since $|A| < |B| \Rightarrow$ at least one element from B will have the same image with another one.
 $\Rightarrow g$ is not injective

Also, f is not surjective because at least one element won't be the image of an element of A .

e) $\begin{array}{l} 1 \rightarrow \\ 2 \rightarrow \\ 3 \rightarrow 2 \end{array} \Bigg| \Rightarrow 6 \text{ functions} \quad \begin{array}{ccc} f(1) < f(2) < f(3) \\ & | & \downarrow \\ & 1 & 2 \end{array}$

6) $\{4, 5, 6, 7, 8, 9, 10\}$

$C_4^3 = 4$

a) $2^7 = 128 - 1 + 1 = 128$

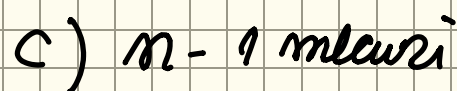
$\begin{array}{c} \downarrow \quad \downarrow \\ \{ \emptyset \} \quad \{ 3 \} \end{array}$

b) $\{1, 2, \dots, 10\} = \{ \quad ? \}$

$$\Rightarrow \text{Total} = 9.7.5.3$$
$$= 63.15$$
$$= 945$$

2 → 1 meci.

$$1. \frac{2^6 - 1}{2 - 1} = 63$$



3 Cu 3, 2, 1

$$n:$$
$$\begin{array}{r} 20 \\ 11 \\ \hline \end{array}$$

3	0	0
2	1	0
1	1	1

2 2 2

9)

B F B F B F B F
F B F B F B F B

B
F B
B F
F B
2 B
3 F
4 2
5 1
6 8
7 7

? ? ? ?

10) $M = \{1, 2, \dots, n\}$

$M = \{1, 2, 3\}$

$A = \{1, 1\}$

B